

Continuous Distributions and the Normal Distribution

STAT 200 – Introductory Statistics

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Today's Objective

After today's lecture, you should:

- Understand the difference between discrete and continuous random variables.
- Know the purpose of the probability density function.
- Be able to prove that the density is not a probability.
- Know the purpose of the cumulative distribution function.
- Be able to calculate probabilities using geometry or the CDF.
- Understand the UNIFORM distribution in terms of its two parameters, sample space, expected value and variance.
- Determine what random variables may follow an EXPONENTIAL distribution.
- Specify the characteristics of an EXPONENTIAL distribution.
- Compare and contrast the EXPONENTIAL with the UNIFORM distribution.
- Calculate the quantiles using the CDF.

Continuous Random Variables

Continuous Random Variables

A **continuous random variable** is a random variable with a sample space consisting of an interval of values.

Examples:

- Student height
- The age of your car
- Time spent at a stoplight
- Distance a baseball goes

Near-Continuous Random Variables

Frequently, a discrete random variable is “essentially” continuous because the spacing between adjacent values is insignificant when compared to the useful data range

Examples:

- Grade point average
- Annual salary
- Gross domestic product (GDP)
- GDP per capita
- Crime rate
- The number of ears of corn grown in Illinois
- The number of COVID-19 cases in a week on campus.

Near-Continuous Random Variables

Why does this matter?

- In terms of analysis, there is *no* difference in how continuous and near-continuous variables are handled. Thus, we will ignore the distinction between near-continuous and continuous.
- Statistical procedures dealing with continuous variables are (usually) identical to those dealing with near-continuous variables.
- As such, the near-continuous are treated as continuous for all intents and purposes.

The UNIFORM Distribution

UNIFORM Distribution

The UNIFORM **distribution** is a continuous distribution that describes random variables whose likelihood of occurring is constant across a specified interval.

If the random variable X has a UNIFORM distribution, we write:

$$X \sim \text{UNIFORM}(a, b)$$

The UNIFORM Distribution

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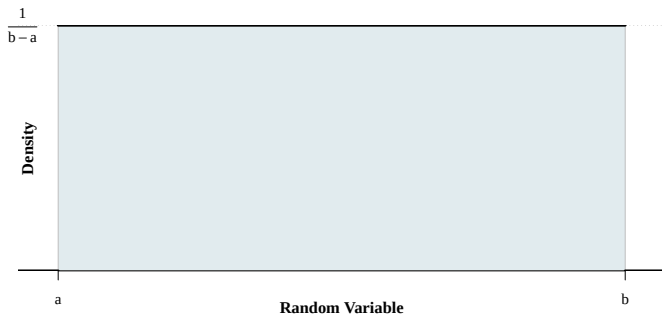
The UNIFORM distribution is characterized by these features:

- It has two parameters: minimum and maximum values (a and b).
- The sample space is $S = [a, b]$.
- The expected value is $\mathbb{E}[X] = \frac{a+b}{2}$.
- The variance is $\mathbb{V}[X] = \frac{(b-a)^2}{12}$.

Probability Density Function of a UNIFORM r.v.

The probability density function for the UNIFORM distribution is given by:

$$f(x; a, b) = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$



Notes on the Probability Density Function

Like the probability mass function for a discrete random variable, the probability density function uniquely defines a continuous distribution.

Note that the probability density function *does not* represent $\mathbb{P}[X = x]$, unlike the probability mass function of a discrete random variable. This can be hard to understand at first!

Technically, $\mathbb{P}[X = x] = 0$ for any specific value of x .

Probabilities are defined as **areas** *under* the density curve.

Example #1: Main and Academy

There is a particularly pesky stoplight between my home and campus in the morning. It regularly cycles among green (for 175 seconds), yellow (5 seconds), and red (180 seconds).

Given that I stop at the light, what is the probability that I wait at most 60 seconds?

Example #1: Main and Academy

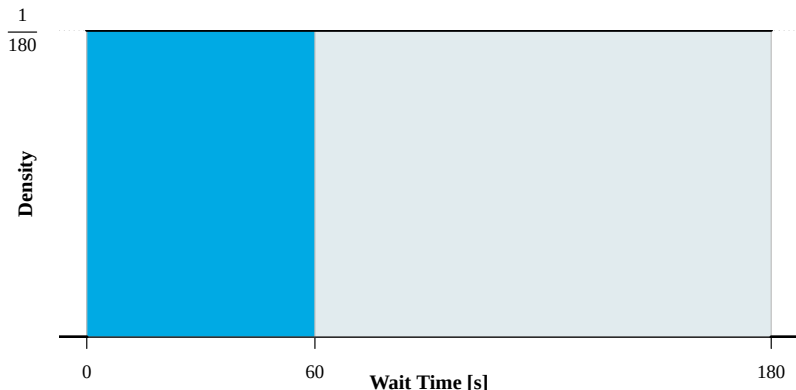
Because the time I wait does not depend on when I get there, and because there is a definite lower- and upper-bound to my wait time (0 and 180 seconds, respectively), the wait time distribution follows a UNIFORM distribution. If we define T as the time (in seconds) I spend waiting at this light, then

$$T \sim \text{UNIFORM}(0, 180)$$

So we need to calculate $\mathbb{P}[T \leq 60]$.

Example #1: Main and Academy

Here is that depicted graphically. The large rectangle is the pdf of $\text{UNIFORM}(0, 180)$, and the dark rectangle is the area of $\mathbb{P}[T \leq 60]$.



Example #1: Main and Academy

Because the pdf of the UNIFORM distribution is just a rectangle, and because areas in pdfs are probabilities, we just need to calculate the area of the region $T \leq 60$.

This is just a rectangle, with area $A = bh$.

$$\begin{aligned}b &= 60 \\h &= \frac{1}{180} \\ \mathbb{P}[X \leq 60] &= \frac{1}{180} \times 60 = \frac{1}{3} = 0.33333 \dots\end{aligned}$$

Thus, the probability of waiting at most one minute at the stoplight on Main and Academy is 33.3%.

Cumulative Distribution Function

Cumulative Distribution Function

The **cumulative distribution function** (CDF) of a probability distribution is defined as

$$F(x) = \mathbb{P}[X \leq x]$$

As we discussed before, this is the first place we've seen a probability when dealing with continuous random variables. That is because this is the only type of probability that can be calculated with continuous distributions.

Cumulative Distribution Function

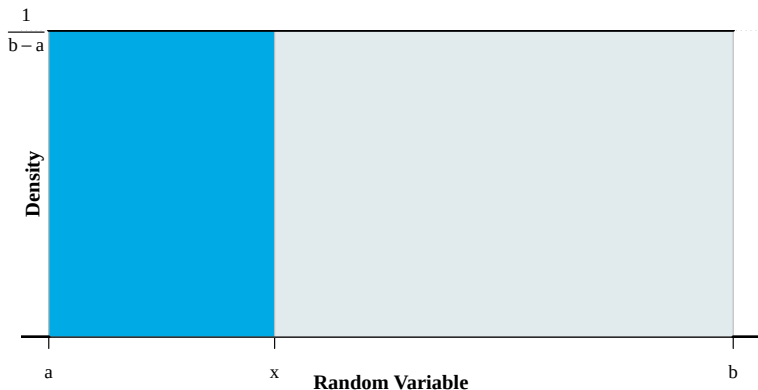
The CDF is frequently difficult to calculate. Usually, it requires integral calculus. Specifically:

$$F(x) = \mathbb{P}[X \leq x] = \int_{-\infty}^x f(t)dt$$

However, for the UNIFORM distribution, we can rely on middle-school geometry.

Cumulative Distribution Function of a UNIFORM r.v.

The cumulative distribution function (CDF) is the function that describes $F(x) = \mathbb{P}[X \leq x]$.



Cumulative Distribution Function of a UNIFORM r.v.

Let $X \sim \text{UNIFORM}(a, b)$.

The probability $\mathbb{P}[X \leq x]$ is the area of the rectangle with height $f(x) = \frac{1}{b-a}$ and width $x - a$.

This means the cumulative distribution function is:

$$\mathbb{P}[X \leq x] = F(x) = \frac{x - a}{b - a}$$

Quantile Function

The CDF starts with a value of the random variable, x , and calculates a cumulative probability, p , such that $\mathbb{P}[X \leq x] = p$.

The quantile function starts with a cumulative probability, p , and calculates the x -value such that $\mathbb{P}[X \leq x_p] = p$.

The two function are inverses of each other, i.e. $Q(p) = F^{-1}(p)$.

Quantile Function of the UNIFORM Distribution

The quantile function for the UNIFORM distribution is

$$p = \frac{x - a}{b - a}$$

$$p(b - a) = x - a$$

$$p(b - a) + a = x$$

Thus, the quantile function for the UNIFORM distribution is

$$Q(p) = p(b - a) + a$$

UNIFORM Distribution in R

In R, there are a set of four functions to calculate various quantities from the UNIFORM distributions

- `dunif(x, min, max)` is the pdf, $f(x) = p$.
- `punif(x, min, max)` is the CDF, $F(x) = \mathbb{P}[X \leq x] = p$.
- `qunif(p, min, max)` is the quantile function, i.e. solving $F(x) = \mathbb{P}[X \leq x] = p$ for x .
- `runif(n, min, max)` generates n random values from a `UNIFORM(min, max)` distribution.

The EXPONENTIAL Distribution

EXPONENTIAL Distribution

The EXPONENTIAL **distribution** is a continuous distribution that describes random variables whose probability of occurring decreases with time. It is frequently used to model “time until” something happens when there is no upper bound.

If the random variable X has a EXPONENTIAL distribution, we write:

$$X \sim \text{EXPONENTIAL}(\lambda)$$

The EXPONENTIAL Distribution

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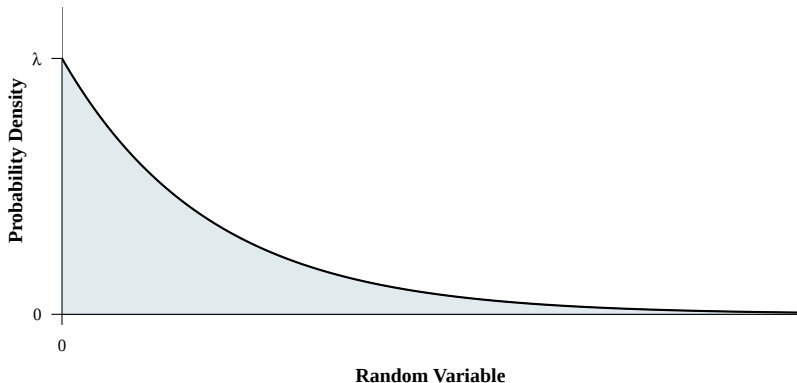
The EXPONENTIAL distribution is characterized by these features:

- It has one parameters: λ , the rate of the event occurring.
- The sample space is $S = (0, \infty)$.
- The expected value is $\mathbb{E}[X] = \frac{1}{\lambda}$.
- The variance is $\mathbb{V}[X] = \frac{1}{\lambda^2}$.

Probability Density Function of a EXPONENTIAL r.v.

The probability density function for the EXPONENTIAL distribution is given by:

$$f(x; a, b) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$



The CDF of the EXPONENTIAL Distribution

The cumulative distribution function of the EXPONENTIAL distribution is:

$$\mathbb{P}[X \leq x] = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases}$$

Proving this isn't too hard, but requires an integral. I'll post the proof alongside these slides, but it's not required.

EXPONENTIAL Distribution in R

In R, there are a set of four functions to calculate various quantities from the EXPONENTIAL distributions

- `dexp(x, rate)` is the pdf, $f(x) = p$.
- `pexp(x, rate)` is the CDF, $F(x) = \mathbb{P}[X \leq x] = p$.
- `qexp(p, rate)` is the quantile function, i.e. solving $F(x) = \mathbb{P}[X \leq x] = p$ for x .
- `rexp(n, rate)` generates n random values from a EXPONENTIAL(`rate`) distribution.

Characteristics of the NORMAL Distribution

Here are some important features of the NORMAL distribution:

- It is named after Carl Friedrich Gauss (1777–1855) who advocated for its use.
- The NORMAL distribution arises from modeling observed randomness (errors) in astronomical and geodetic data.
- It arises in variables that have a specific expected value, but demonstrate minor random variation.
- It has two parameters: μ (expected value) and σ (standard deviation).
- It is the most important distribution in statistics because of the Central Limit Theorem.

Carl Friedrich Gauss (1777–1855)

Carl Friedrich Gauss is one of the most famous and important mathematicians ever:



https://en.wikipedia.org/wiki/List_of_things_named_after_Carl_Friedrich_Gauss

Probability Density Function of the NORMAL

If a random variable follows a NORMAL distribution with mean μ and standard deviation σ , i.e. $X \sim \text{NORMAL}(\mu, \sigma)$, the the probability density function (pdf) is:

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\sigma^2}} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}$$

Standard NORMAL Distribution

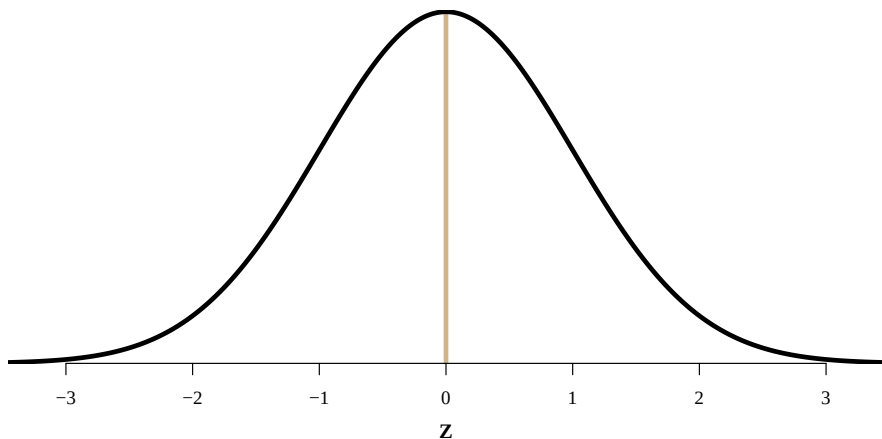
This function depends on the values of μ and σ . If we standardize X , meaning take the z -score for all the values, we get the **Standard Normal Distribution**:

$$Z := \frac{X - \mu}{\sigma} \sim \text{NORMAL}(0, 1)$$

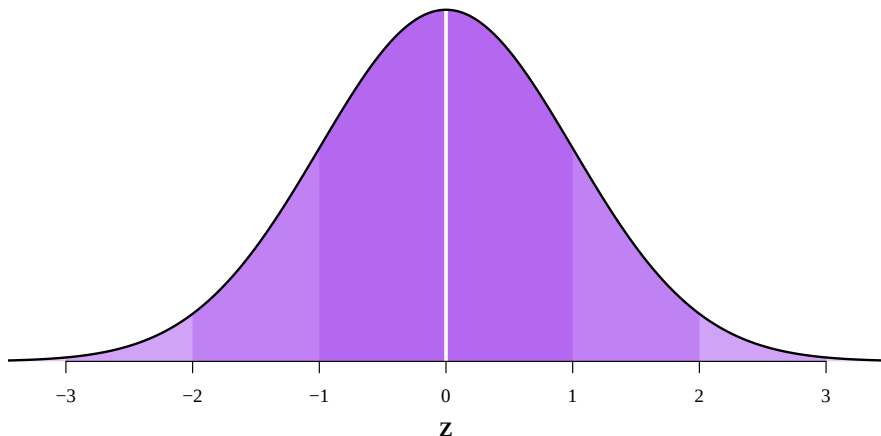
This has mean 0 and standard deviation 1.

If you've ever used a Z -table, this is the NORMAL distribution that it describes.

Shape of the Standard NORMAL Distribution



Shape of the Standard NORMAL Distribution



Cumulative Distribution Function of the NORMAL

Recall that the NORMAL distribution is a continuous distribution, so it's defined by its cumulative distribution function, and $\mathbb{P}[X = x] = 0$ for any x .

The cumulative distribution function of the NORMAL distribution is:

$$F(x; \mu, \sigma) = \mathbb{P}[X \leq x] = \int_{-\infty}^x \frac{1}{\sqrt{2\sigma^2}} \exp \left\{ -\frac{(t - \mu)^2}{2\sigma^2} \right\} dt$$

This integral has no form, so we either need to use tables of CDF of the standard NORMAL distribution, Z , typically denoted $\Phi(x)$, or use your computer, which is faster, easier and more accurate.

NORMAL Distribution in R

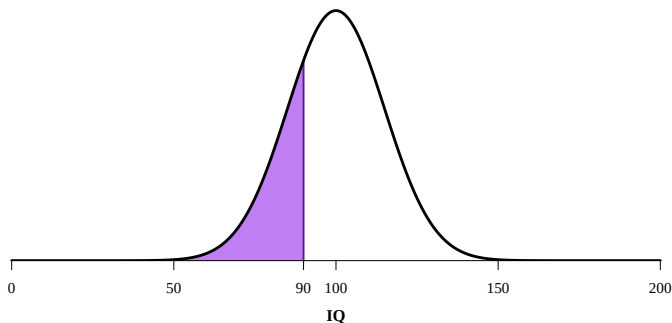
In R, there are a set of four functions to calculate various quantities from the NORMAL distributions

- `dnorm(x, mean, sd)` is the pdf, $f(x) = p$.
- `pnorm(x, mean, sd)` is the CDF, $F(x) = \mathbb{P}[X \leq x] = p$.
- `qnorm(p, mean, sd)` is the quantile function, i.e. solving $F(x) = \mathbb{P}[X \leq x] = p$ for x .
- `rnorm(n, mean, sd)` generates n random values from a `NORMAL(mean, sd)` distribution.

Example #2: Intelligence Quotient (IQ)

Example

By design, intelligence quotient (IQ) measures in the United States follow a NORMAL distribution with $\mu = 100$ and $\sigma = 15$. What proportion of Americans have an IQ score less than 90?



Example #2: Intelligence Quotient (IQ)

Solution: We are asked to calculate p such that

$$\mathbb{P}[X \leq 90] = p$$

given that $X \sim \text{NORMAL}(100, 15)$.

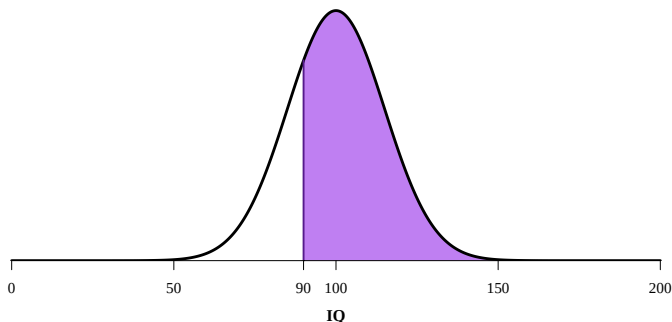
This can be calculated in R with `pnorm(90, mean = 100, sd = 15)`.

Thus, the proportion of people in the United States with IQ scores less than 90 is approximately 0.2524925, which is about 25%.

Example #3: Intelligence Quotient (IQ)

Example

By design, intelligence quotient (IQ) measures in the United States follow a NORMAL distribution with $\mu = 100$ and $\sigma = 15$. What proportion of Americans have an IQ score above 90?



Example #3: Intelligence Quotient (IQ)

Solution: We are asked to calculate p such that

$$\mathbb{P}[X > 90] = p$$

given that $X \sim \text{NORMAL}(100, 15)$.

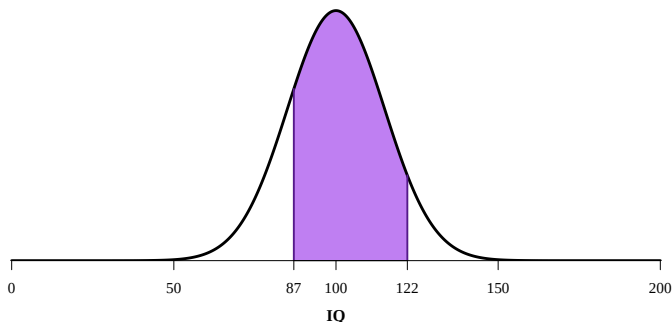
This can be calculated in R with `1 - pnorm(90, mean = 100, sd = 15)`.

Thus, the proportion of people in the United States with IQ scores above 90 is approximately 0.7475075, which is about 75%.

Example #4: Intelligence Quotient (IQ)

Example

By design, intelligence quotient (IQ) measures in the United States follow a NORMAL distribution with $\mu = 100$ and $\sigma = 15$. What proportion of Americans have an IQ score between 87 and 122?



Example #4: Intelligence Quotient (IQ)

Solution: We are asked to calculate p such that

$$\mathbb{P}[87 < X < 122] = p$$

given that $X \sim \text{NORMAL}(100, 15)$.

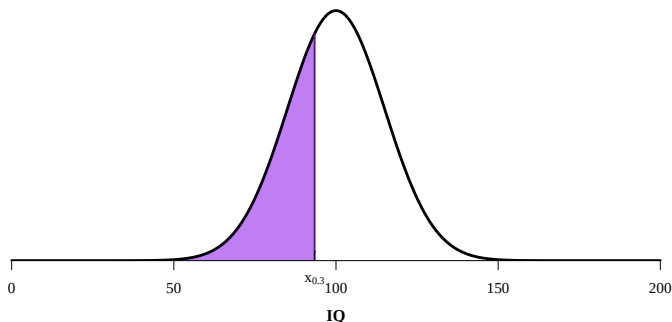
This can be calculated in R with `pnorm(122, mean = 100, sd = 15)`
- `pnorm(87, mean = 100, sd = 15)`.

Thus, the proportion of people in the United States with IQ scores between 87 and 122 is approximately 0.7357043, which is about 75%.

Example #5: Intelligence Quotient (IQ)

Example

By design, intelligence quotient (IQ) measures in the United States follow a NORMAL distribution with $\mu = 100$ and $\sigma = 15$. What IQ value separates the lower 30% from the upper 70%?



Example #5: Intelligence Quotient (IQ)

Solution: We are asked to calculate p such that

$$\mathbb{P}[X < x_p] = 0.30$$

given that $X \sim \text{NORMAL}(100, 15)$.

This can be calculated in R with `qnorm(0.30, mean = 100, sd = 15)`.

Thus, approximately 30% of Americans have an IQ score less than 92.13 and approximately 70% of Americans have an IQ score of more than 92.13.